**Semester Project**

# Training a High-Accuracy Markerless Salamander Pose Estimator with X-Ray Radiography


Author:

**Gal Pascual**

Supervisors:

Chuanfang Ning
Qyuan Fu


**June 2025**

# Contents


**1 Introduction** **3**
- 1.1 Motivation . . . 3
- 1.2 Report Outline . . . 3

**2 State of the Art** **3**

**3 Planned vs. Implemented Pipeline** **4**
- 3.1 Initial Workflow . . . 4
- 3.2 Implemented Workflow . . . 4

**4 Data** **5**
- 4.1 Acquisition Setup . . . 5
- 4.2 Recording Sessions . . . 5
- 4.3 Subjects . . . 6
- 4.4 Marker positions and Anatomical Landmarks . . . 8
- 4.5 Terrain . . . 9

**5 Pre-Processing** **9**

**6 Semi-Automatic Labelling** **11**
- 6.1 Kalman filter . . . 11
  - 6.1.1 System Model . . . 11
  - 6.1.2 Process Noise . . . 11
  - 6.1.3 Measurements . . . 12
  - 6.1.4 Likelihood Function . . . 12
- 6.2 Noise Covariance Modulation . . . 13
- 6.3 Manual Overrides . . . 16
- 6.4 Reprojection . . . 16
- 6.5 Observed Limitations . . . 16

**7 DeepLabCut training** **18**
- 7.1 Dataset . . . 18
- 7.2 Symmetry Augmentation . . . 19
- 7.3 Cropping . . . 19
- 7.4 Hyper-parameter Selection and Library Adjustments . . . 19

**8 Results** **20**
- 8.1 Evaluation Metrics . . . 20
- 8.2 Training Histories . . . 20
- 8.3 Validation Set Results . . . 21
- 8.4 Test Results – Complex-Terrain Sequence . . . 22

**9 Conclusions and Future Improvements** **23**
- 9.1 Data Acquisition (X-ray & Optical) . . . 23
- 9.2 Image Pre-processing . . . 24
- 9.3 Semi-Automatic Tracking Script . . . 24
- 9.4 Deep-Learning Model Training . . . 24

# 1 Introduction

## 1.1 Motivation

Accurate three-dimensional pose information is fundamental for quantitative analyses of locomotion and behavior in vertebrates. Precise understanding of salamander locomotion specifically contributes significantly to the study of neuromechanical control mechanisms in vertebrates and has critical implications in biomimetic robotics. The present project focuses on the Salamander *Pleurodeles*, whose elongated body and undulatory gait pose a demanding benchmark for vision-based tracking systems.

The original objective was to train a deep-learning framework capable of inferring the salamander's 3D pose from synchronized multi-view optical recordings. To obtain geometrically precise ground-truth data, radiopaque lead markers were surgically implanted at key skeletal landmarks. Biplanar X-ray cinematography was expected to deliver exact 3D marker coordinates, which would then be re-projected onto optical images, supplying automatically labeled training frames for a pose-estimation network such as DeepLabCut. Due to practical constraints, modifications to the initial workflow were necessary; these adaptations and their rationale are detailed in Section 3.

## 1.2 Report Outline

The remainder of this document is organised as follows:

- section 2 (State of the Art) reviews the main techniques currently used for high-accuracy animal pose estimation, ranging from marker-less (DeepLabCut, DLC-3D, Anipose) to marker-based XROMM approaches.
- section 3 (Planned vs. Implemented Pipeline) contrasts the initially planned four-stage pipeline with the procedure ultimately implemented, explaining each deviation and its rationale.
- section 4 (Data) describes the experimental setup, recording sessions, salamander subjects, marker configurations, and terrain conditions.
- section 5 (Pre-Processing) details the image pre-processing chain that enhances fluoroscopic marker visibility before tracking.
- section 6 (Semi-Automatic Labelling) explains the semi-automatic labelling workflow, including the Kalman filter, noise-covariance modulation, manual corrections, and observed limitations.
- section 7 (DeepLabCut training) presents the DeepLabCut training strategy: dataset construction, symmetry augmentation, cropping, and hyper-parameter tuning.
- section 8 (Results) reports quantitative metrics and qualitative examples that assess the model's accuracy and generalisation.
- section 9 (Conclusions and Future Improvements) summarises the findings and outlines immediate avenues for improving data acquisition, tracking automation, and deep-learning training.

# 2 State of the Art

**DeepLabCut (2-D).** *DeepLabCut* (DLC) was introduced in 2018 by Mathis *et al.* as an open-source, marker-less pose-estimation toolbox that fine-tunes an ImageNet-pre-trained ResNet-50 with as few as 50–200 labelled frames. Its data-augmentation pipeline, two-stage score-map refinement, and simple API have made DLC the de-facto standard for animal kinematics.

**DLC-3D.** The official *DLC-3D* extension triangulates 2-D key-points from synchronised cameras using a straightforward pinhole calibration. With careful camera geometry, DLC-3D achieves sub-millimetre accuracy on freely moving rodents, fish, and lizards, offering a lightweight alternative to reflective-marker motion-capture systems.

**Anipose.** *Anipose* builds on DLC outputs but adds automated multi-view calibration, temporal filtering, and kinematic constraints. The toolbox can process entire experiments end-to-end—from raw videos to 3-D joint angles—using a lighter U-Net backbone for higher frame-rate throughput than vanilla DLC.

**XROMM with radio-opaque markers.** X-ray Reconstruction of Moving Morphology (XROMM) remains the gold standard when sub-pixel skeletal accuracy is required. Radio-opaque bead markers are implanted surgically; their 3-D trajectories, recovered by bi-planar fluoroscopy, reach $< 0.1$ mm precision but at the cost of invasive surgery and heavy hardware. Marker-based XROMM is often used as ground truth to validate marker-less methods like DLC-3D and Anipose.

**Summary.** DLC dominates marker-less 2-D tracking with minimal labelling effort; DLC-3D and Anipose extend this power to calibrated multi-view setups, while XROMM provides the highest physical fidelity when implants and specialised X-ray equipment are acceptable. Together, these tools form the current ecosystem for high-accuracy animal pose estimation. What remains missing, however, is a pipeline that eliminates the need for manual annotations altogether—combining the geometric precision of X-ray projections with automatic labelling to accelerate the process and improve landmark accuracy. This project addresses that gap.

# 3 Planned vs. Implemented Pipeline

## 3.1 Initial Workflow

The initial plan relied entirely on X–ray guidance to generate ground–truth labels. After enhancing marker contrast, a semi–automatic Kalman tracking script was expected to follow all radiopaque beads over an *entire* X–ray sequence (approximately 1,500 frames). Triangulated 3–D coordinates would then be reprojected onto three synchronised optical views, and the full set of projected coordinates would serve as dense labels for training DeepLabCut.

## 3.2 Implemented Workflow

The implemented procedure retained the core logic of the initial pipeline—pre-processing, semi-automatic tracking, triangulation, reprojection and training—but required targeted manual corrections and additional manual annotations due to practical constraints:

1. **XR-guided annotation.** An initial X-ray/optical sequence was processed following the planned workflow: pre-processing for enhanced marker contrast, semi-automatic marker tracking in X-ray images, triangulation, and reprojection onto optical views. Representative frames were then manually reviewed: clearly misaligned markers (such as limb joints, tail vertebrae, or occluded parts) were corrected or removed, while visually ambiguous markers (e.g., upper spinal vertebrae) generally remained unchanged.

2. **Fully manual annotation of optical views.** Because of significant overhead in tuning the tracking script across multiple sequences, additional sequences were annotated entirely by hand to efficiently diversify the training data without further automation effort.

3. **Dataset combination and model training.** XR-guided annotations and fully manual annotations were merged into a unified training set, combining the geometric precision provided by X-ray

data for difficult-to-identify landmarks with manual annotations enhancing dataset diversity and practicality.

Ideally, the entire dataset would have been annotated via XR-guided reprojection, but manual labelling provided a timely alternative for increasing generalisation across diverse sequences. Subsequent sections detail each step of the pipeline—pre-processing, tracking, reprojection, and manual correction—along with the specific dataset used for model training.

# 4 Data

## 4.1 Acquisition Setup

The dataset combines optical and fluoroscopic recordings. Three synchronised grayscale cameras capture the external appearance of the animal, while two high-speed X-ray emitters and intensifiers provide biplanar projections of the implanted radiopaque markers. Devices acquired data at a 700 fps bandwidth and stored it in the local flash memory. When the flash capacity is filled (2500 frames), the data is copied into non-volatile media.

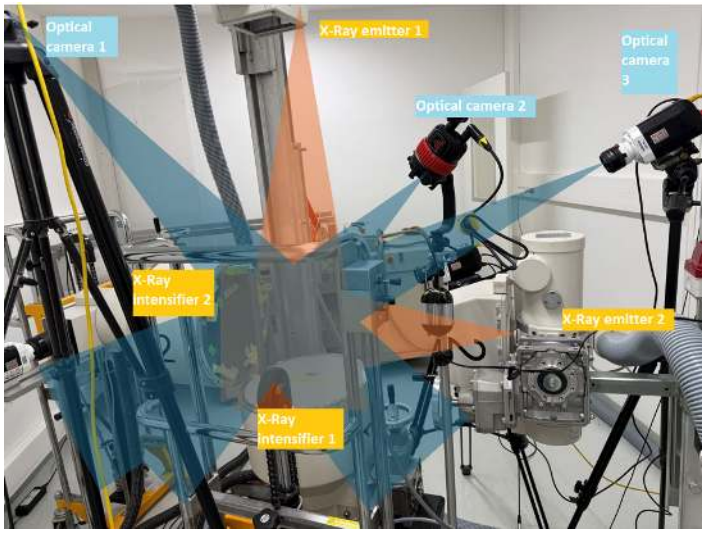


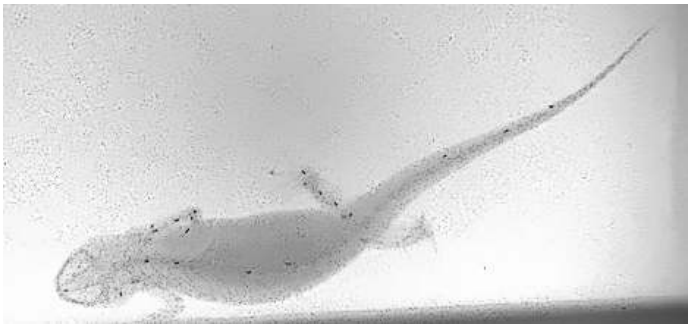

Figure 1: Experimental setup of the 20th of February

## 4.2 Recording Sessions

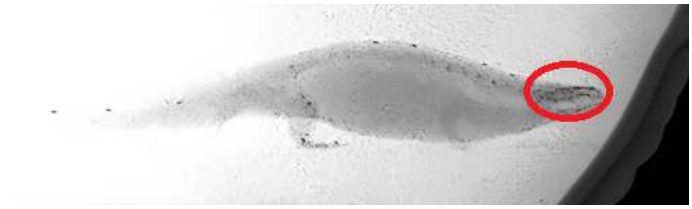
Two experimental campaigns were performed with different X-Ray views placements(see Figure 2, pre-processed for enhanced visibility).

**20 February 2025** X-ray views were arranged dorsally (top view) and laterally (side view).Although the dorsal projection offers excellent marker contrast, the lateral beam traverses a greater tissue thickness, producing blurred contours and frequent superposition between markers and between markers and bone (Figure 2a, Figure 2b). Additional constraints affecting image quality and marker visibility include a limited field of view, glass thickness in the detector assembly, suboptimal X-ray contrast for certain anatomical regions, occasional missing markers, and unintended extra markers (the markers moved post surgery).

**15 April 2025** To reduce superposition, the lateral unit was repositioned to an oblique angle (45° off vertical). Contrast improves, but the oblique view demonstrated an uneven fluence field due to the broken blender of one of the two X-ray emitters: posterior sections of the spine are systematically darker and several caudal markers occasionally drop below the detection threshold (Figure 2c).



(a) Dorsal view (20 Feb)



(b) Lateral view (20 Feb), frequent overlap in the red region



(c) Oblique view (15 Apr), low contrast in the red region

Figure 2: Pre-processed X-ray projections illustrating marker visibility and superposition in the two recording sessions.

## 4.3 Subjects

Initially, four salamanders (Pleurodeles 1–4; see Figure 3) were recorded during the February session. **Pleurodeles 1** was retained as the reference animal because it carries the most complete and correctly placed marker set, enabling a three-segment limb model (girdle–femur, femur–tibia, tibia–pes). **Pleurodeles 2**, by contrast, lacks the distal tibia–pes markers on its hind limbs, so only a two-segment representation is possible; this reduced limb resolution applies only to Pleurodeles 2.

The main defects for each subject are summarised below.

Table 1: Pleurodeles 1 — marker anomalies

| **Type** | **Markers concerned** |
|---|---|
| Missing | Left head, left hind claw, spine 5 |
| Misplaced | Left hind femur |

Table 2: Pleurodeles 2 — marker anomalies

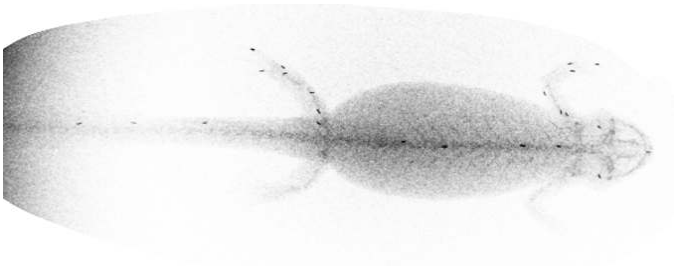
| **Type** | **Markers concerned** |
|---|---|
| Missing | Head tip, last caudal vertebra, both LH claws, LF claw |
| Misplaced | – |

Table 3: Pleurodeles 3 — marker anomalies

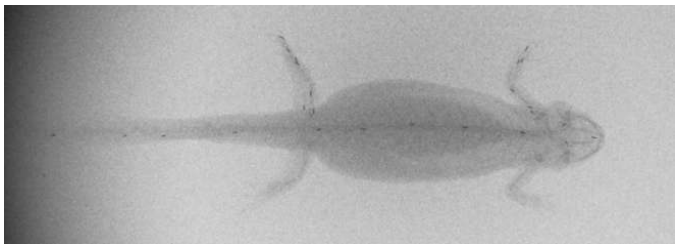
| **Type** | **Markers concerned** |
|---|---|
| Missing | All head markers, last caudal vertebra, both LH claws |
| Misplaced | – |

Table 4: Pleurodeles 4 — marker anomalies

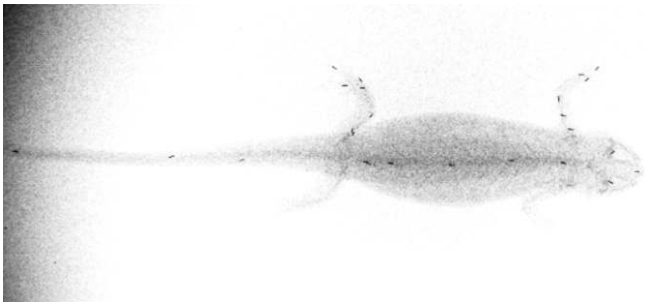
| **Type** | **Markers concerned** |
|---|---|
| Missing | LF claw, last caudal vertebra, both LH claws |
| Misplaced | LF femur |



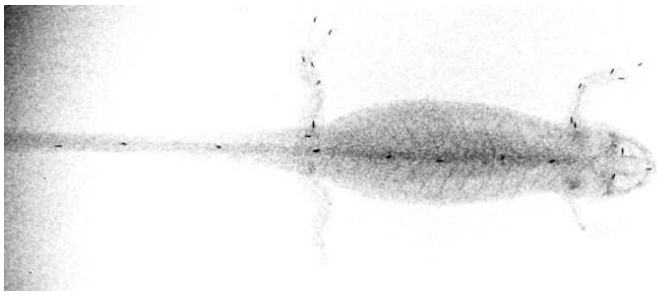
(a) Pleurodeles 1



(b) Pleurodeles 2



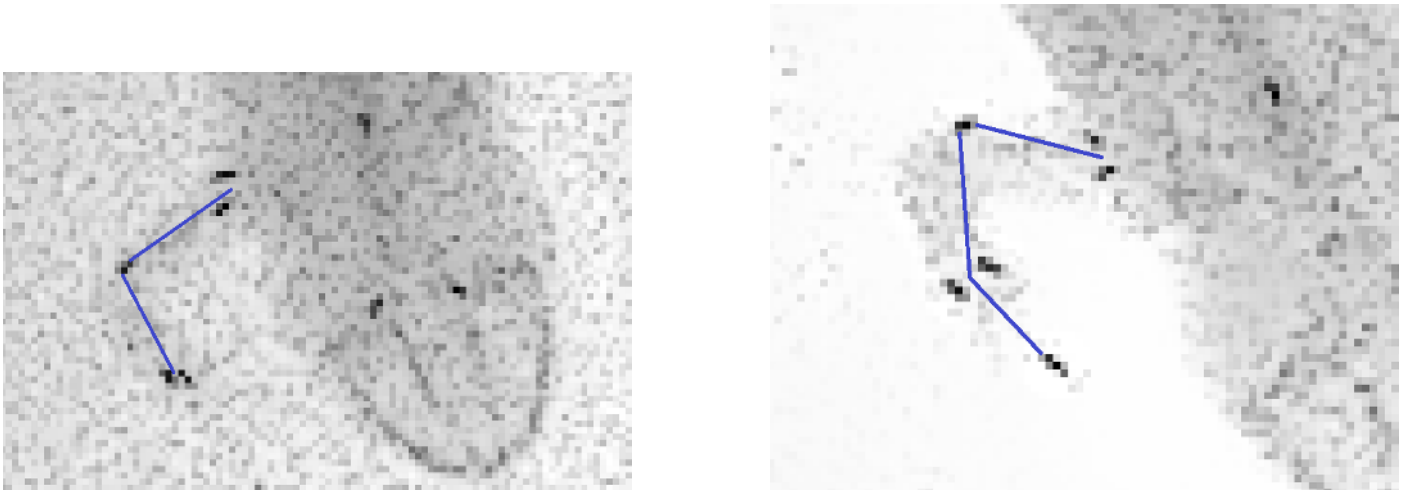
(c) Pleurodeles 3



(d) Pleurodeles 4

Figure 3: Dorsal and lateral X-ray snapshots of the four subjects, annotated with the detected marker defects.

Due to time constraints, the X–ray reprojection pipeline was ultimately applied only to Sequence 6 of Pleurodeles 1 from February session, which offered a big number of frames with all markers simultaneously visible in both X–ray views. Additional limb markers were manually annotated on 4 *Pleurodeles* walking on flat terrains from the April session. Each salamander performed approximately ten locomotion trials on flat terrain.



(a) Pleurodeles 2: two-segment limb

(b) Pleurodeles 1: three-segment limb

Figure 4: Marker configurations illustrating differences in limb segmentation capability.

## 4.4 Marker positions and Anatomical Landmarks

Due to ethical considerations, radiopaque markers were implanted exclusively on one side of the salamander. Consequently, the dataset is augmented by horizontally mirroring both the images and the corresponding labels.

The anatomical landmarks predicted by the pose-estimation model include:

- Head: tip, left and right sides
- Spine: vertebrae 1 through 8
- Limbs (left and right; front and hind):
  - Shoulder/Hip girdle
  - Femur–tibia (FeTi) joint
  - Tibia–pes (TiPe) joint
  - Claw extremity

Markers did not always coincide exactly with the targeted anatomical locations. Thus, during manual corrections, slight positional corrections were applied—particularly at the femur–tibia joints. In cases where multiple markers corresponded to a single anatomical landmark (e.g., girdle or TiPe joint), their positions were averaged to generate the final annotation point.

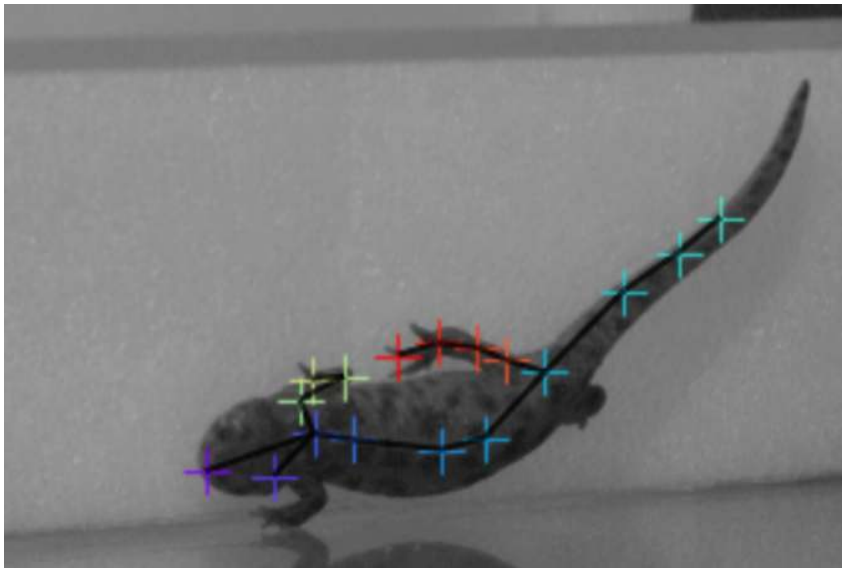
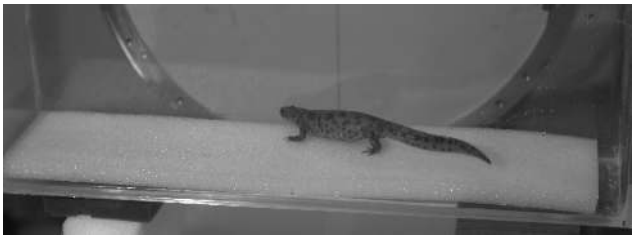


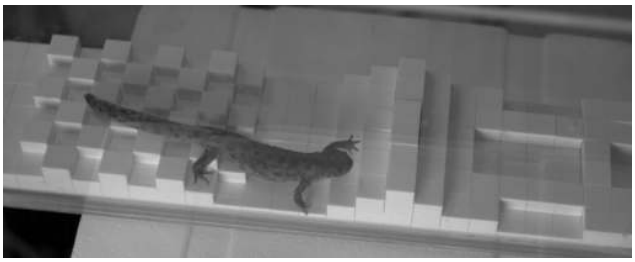
Figure 5: Half-annotated salamander image with anatomical landmarks.

### 4.5 Terrain

Each session contains level-ground locomotion and traversal of uneven terrain (Figure 6). Because rough terrain severely degrades marker visibility in both modalities, only the level-ground sequences enter the training and validation sets. Uneven-terrain sequences are preserved solely for qualitative tests of model generalisation.



(a) Flat (level) terrain



(b) Complex (uneven) terrain

Figure 6: Examples of (a) flat terrain used for training/validation and (b) complex terrain reserved for qualitative generalisation tests.

## 5 Pre-Processing

The raw fluoroscopic images are highly noisy (see Figure 8). To enhance marker visibility, the following pipeline is applied:

1. **Exposure compensation** to normalize the global brightness.
2. **Gamma correction** for grey-level re-mapping.
3. **Local-contrast enhancement** to sharpen faint edges.
4. **Unsharp masking** to emphasize high-frequency details.
5. **Non-local means filtering** to suppress residual speckle noise while preserving marker edges.

**Note:** After completing the main experiments, it became evident that the denoising filter would be more effective if applied as the *first* step, prior to any other processing. This adjustment would likely preserve more fine-scale details by suppressing noise before contrast enhancement. However, this

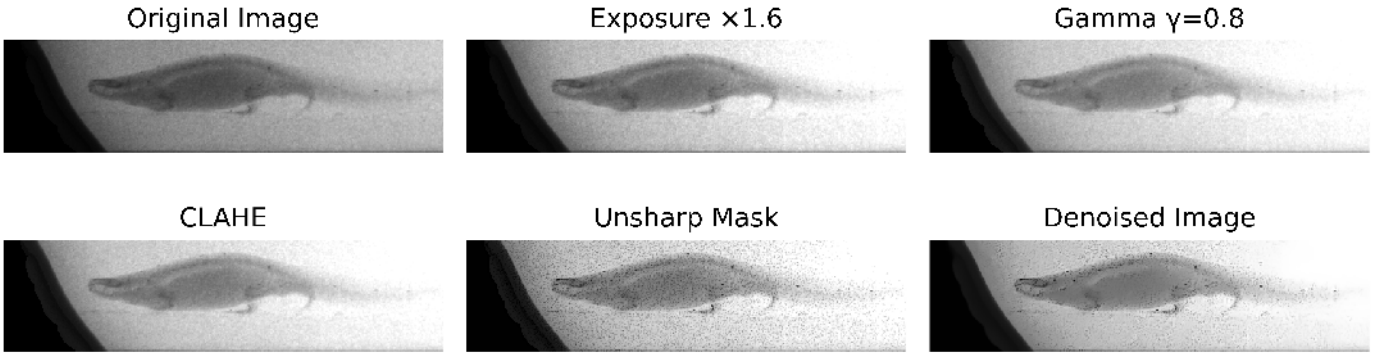
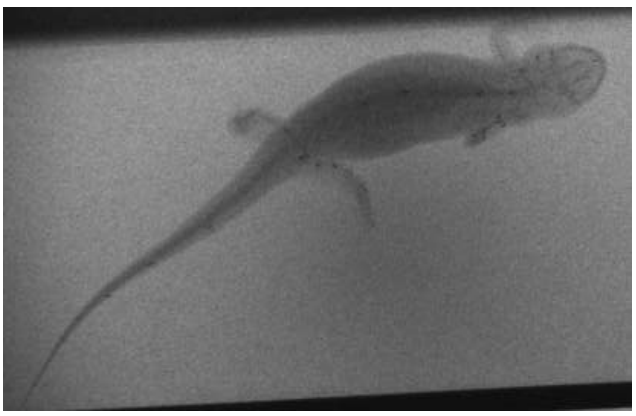
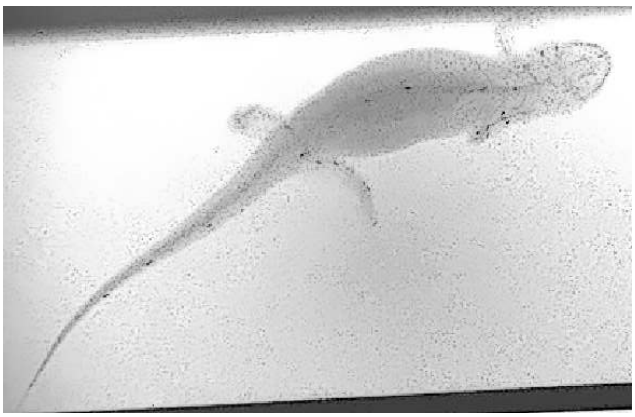


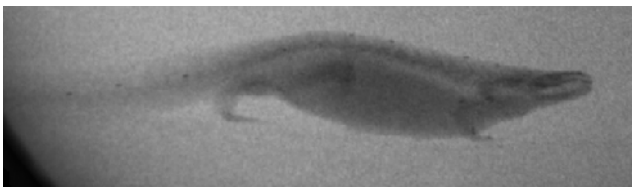

Figure 7: Pre-processing pipeline



(a) Raw dorsal view



(b) Processed dorsal view



(c) Raw lateral view



(d) Processed lateral view

Figure 8: Comparison of raw and pre-processed fluoroscopic images in dorsal and lateral views.

modification was identified too late and was not implemented in the present pipeline. All parameters are tuned per recording, because illumination and beam intensity are not perfectly constant across sessions.

# 6 Semi-Automatic Labelling

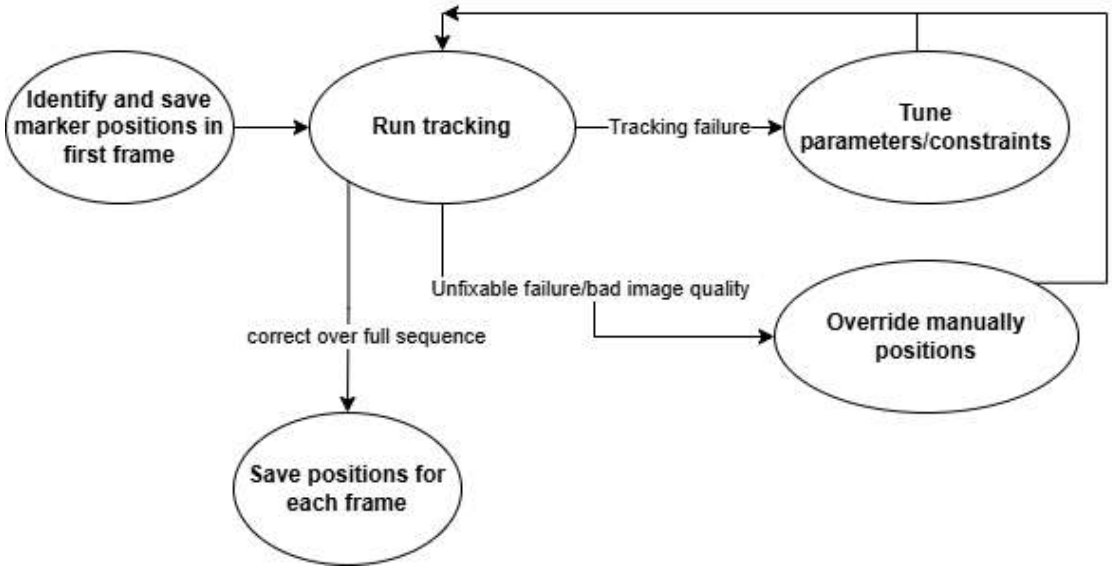



Figure 9: Pipeline to track marker's positions

## 6.1 Kalman filter

Fluoroscopic image sequences are annotated with an independent Kalman filter per view. Each marker is described by a four-dimensional constant-velocity state vector

$$\mathbf{x} = \begin{bmatrix} x & y & u_x & u_y \end{bmatrix}^{\mathsf{T}},$$

where $(x, y)$ denotes the image location and $(u_x, u_y)$ the instantaneous velocity.

### 6.1.1 System Model

The discrete-time dynamics with sampling period $\Delta t$ are given by

$$\mathbf{x}_{k+1} = \mathbf{F}\,\mathbf{x}_k + \mathbf{w}_k, \qquad \mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ models acceleration that is not captured by the constant-velocity assumption.

### 6.1.2 Process Noise

With white, axis–independent acceleration $a_x, a_y \sim \mathcal{N}(0, \sigma)$ the continuous model is $\dot{x} = v_x,\ \dot{v}_x = a_x$ (idem for $y$). Discretising over $\Delta t$ gives

$$\mathbf{x}_{k+1} = \underbrace{\begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{F}} \mathbf{x}_k + \underbrace{\begin{bmatrix} \frac{\Delta t^2}{2} & 0 \\ 0 & \frac{\Delta t^2}{2} \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}}_{\mathbf{G}} \begin{bmatrix} a_{x,k} \\ a_{y,k} \end{bmatrix}.$$

Because $\mathrm{Cov}(a_{x,k}, a_{y,k}) = \sigma \mathbf{I}_2$, the discrete process noise is $\mathbf{Q} = \mathbf{G}\,\sigma \mathbf{I}_2\,\mathbf{G}^{\top}$, i.e.

$$\mathbf{Q} = \sigma \begin{bmatrix} \frac{\Delta t^4}{4} & 0 & \frac{\Delta t^3}{2} & 0 \\ 0 & \frac{\Delta t^4}{4} & 0 & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & 0 & \Delta t^2 & 0 \\ 0 & \frac{\Delta t^3}{2} & 0 & \Delta t^2 \end{bmatrix}$$

where the $\frac{\Delta t^4}{4}$ terms come from double integration of acceleration to position, the $\Delta t^2$ terms from single integration to velocity, and the $\frac{\Delta t^3}{2}$ cross terms from their coupling.

### 6.1.3 Measurements

For each marker the Kalman predictor supplies a location $\hat{\mathbf{x}}_{k+1} = (\hat{x}, \hat{y})^\mathsf{T}$ in the next frame. Around that point we examine a disc of radius three pixels and retain the $N = 7$ darkest pixels These pixels form one or more 4-connected components, each of which becomes a *candidate measurement*.

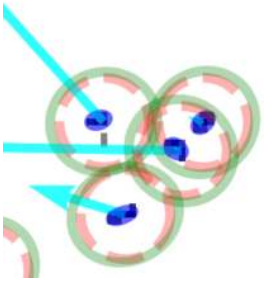


Figure 10: Every connected component is a potential measurement candidate, the dark blue color indicates the selected measurement. The red and green ellipses represent respectively, the previous and actual estimated position.

**Position and Covariance.** For a connected component $\mathcal{P} = \{\mathbf{p}_i\}_{i=1}^n$ with intensities $\{I_i\}$ we define

$$\boldsymbol{\mu} = \frac{\sum_{i=1}^n I_i \, \mathbf{p}_i}{\sum_{i=1}^n I_i}, \qquad \boldsymbol{\Sigma} = \frac{\sum_{i=1}^n I_i \, (\mathbf{p}_i - \boldsymbol{\mu})(\mathbf{p}_i - \boldsymbol{\mu})^\mathsf{T}}{\sum_{i=1}^n I_i}.$$

To remove the scale bias introduced by large blobs we normalise the covariance $\boldsymbol{\Sigma} \leftarrow |\boldsymbol{\Sigma}|^{-1/2}\boldsymbol{\Sigma}$, so that every measurement ellipse has unit area regardless of blob size.

**Hypothesis Branching.** Each marker spawns as many *hypotheses* as there are candidate measurements in its search area; every hypothesis is updated with a distinct $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ pair. After all markers have branched, the algorithm must choose one hypothesis per marker that maximises the global likelihood.

**Combinatorial Search.** The likelihood of a hypothesis depends on the positions of its spatial neighbours, yet the interaction range is short. Instead of evaluating the combinatorial explosion of all markers at once, we cluster the previous positions with DBSCAN ($\varepsilon = 10$ pixels, MINPTS$= 1$). Within each cluster we enumerate every combination of hypotheses, compute the product of their individual likelihoods, and select the most probable configuration. The global solution is obtained as the union of the per-cluster maxima, which is exact because there isn't any inter-cluster interactions.

### 6.1.4 Likelihood Function

For every candidate measurement, the Kalman filter performs a virtual update and returns the innovation

$$\mathbf{y} = \mathbf{z} - \mathbf{H}\hat{\mathbf{x}}, \qquad \mathbf{S} = \mathbf{H}\hat{\mathbf{P}}\mathbf{H}^\mathsf{T} + \mathbf{R},$$

where $\hat{\mathbf{x}}, \hat{\mathbf{P}}$ are the predicted state and covariance, $\mathbf{z}$ the measurement in image coordinates, and $\mathbf{R}$ the image–noise covariance.

The likelihood of a marker hypothesis is the product of four factors:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{innov}}}_{\text{innovation}} \cdot \underbrace{C_{\text{dist}}}_{\text{distance constraint}} \cdot \underbrace{C_{\text{sup}}}_{\text{superposition penalty}} \cdot \underbrace{W_{\text{int}}}_{\text{blob intensity}}.$$

**1. Innovation term** With $d = 2$ image dimensions,

$$\mathcal{L}_{\text{innov}} = (2\pi)^{-d/2}|\mathbf{S}|^{-1/2}\exp\!\big(-\tfrac{1}{2}\,\mathbf{y}^{\mathsf{T}}\mathbf{S}^{-1}\mathbf{y}\big).$$

**2. Binary distance constraint** Some marker pairs $(i, j)$ must remain within $[d^{ij}_{\min}, d^{ij}_{\max}]$ in the image. The factor

$$C_{\text{dist}} = \prod_{(i,j)} \mathbf{1}\big(d^{ij}_{\min} \le \|\mathbf{z}_i - \mathbf{z}_j\| \le d^{ij}_{\max}\big)$$

equals 1 when all distances are valid and 0 otherwise.

**3. Superposition penalty** If two hypotheses belonging to the *same* cluster use the *same* measurement, their likelihood is multiplied by

$$C_{\text{sup}} = \alpha_{ij}, \qquad 0.01 \le \alpha_{ij} < 1.$$

By default $\alpha_{ij} = 0.01$, ensuring a strong penalty. The coefficient can be overridden for specific pairs:

- Pairs that almost never overlap keep the default, effectively excluding the coincidence.
- Pairs that frequently overlap receive a value close to 1 so that shared measurements remain plausible, yet still slightly below 1 to encourage separation when the blobs diverge.

All $\alpha_{ij}$ are tuned empirically.

**4. Blob-intensity weight** Let $\mathcal{P}$ be the pixel set of the connected component and $I(p) \in [0, 255]$ the raw 8-bit intensity (255 = darkest). The weight

$$W_{\text{int}} = \sum_{p\in\mathcal{P}} I(p)$$

assigns higher likelihood to larger and darker blobs, reflecting the expected appearance of well-focused markers.

## 6.2 Noise Covariance Modulation

In our constant–velocity Kalman model, unmodelled accelerations are accounted for by the process-noise covariance

$$\mathbf{Q} = \sigma\,\mathbf{Q}_0,$$

where $\mathbf{Q}_0$ is the base covariance and $\sigma$ scales the trust between model and measurements. For spinal markers—whose true accelerations are mild—we use $\sigma \approx 10^7$, so that the filter relies predominantly on the predicted motion. Limb markers, by contrast, undergo rapid accelerations; here we set $\sigma \approx 10^9$ to allow the filter to track sudden changes.

However, during brief marker superpositions—or whenever two markers come extremely close—an overly high $\sigma$ causes the filter to overweight noisy measurements, lose its velocity estimate, and fail to re-separate the tracks. Even without full overlap, if one marker is slightly larger or darker than its neighbor, the smaller marker may capture only pixels from the larger/darker marker, leading to a merged detection. To prevent this, we apply a dynamic reduction of $\sigma$ whenever any two markers approach closer than a threshold distance $\varepsilon$ (typically 5 px). Specifically, if $\|\mathbf{z}_i - \mathbf{z}_j\| < \varepsilon$, we replace

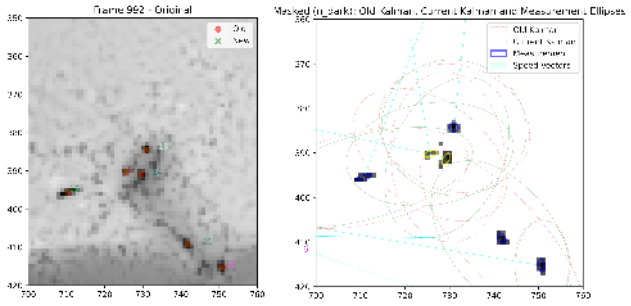
$$\sigma \;\longleftarrow\; \frac{\sigma}{\kappa},$$

with $\kappa \approx 10^3$. This temporary downscaling shifts confidence back to the motion model, allowing the markers to diverge smoothly without label swaps.

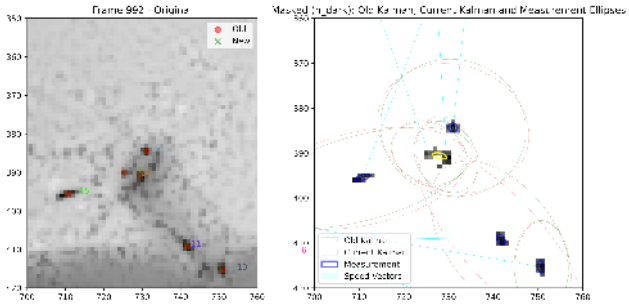
**Implementation.** The reduction of $\sigma$ is not automatic. Instead, a simple configuration file specifies which marker pairs to monitor and by what factor to divide $\sigma$ when they come within a threshold distance. Any marker pairs not listed remain at the default high $\sigma$.

### Qualitative Illustration

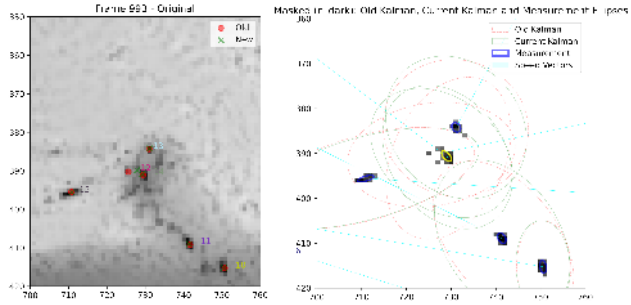
The following shows five consecutive frames (992–996), each in two columns: left without covariance modulation and right with modulation enabled. When the two limb markers come within 5px, dividing $\sigma$ by 1000 prevents the "closest-pixels" selection from shifting one marker far from its true location and losing the track.



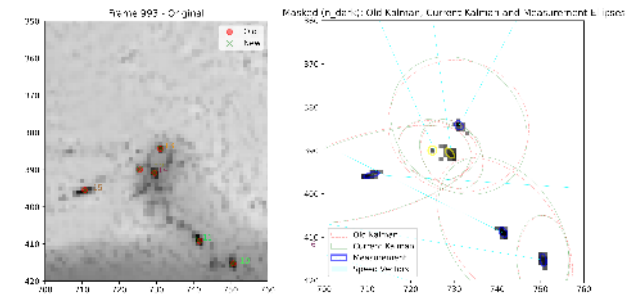
(a) Frame 992 (no mod.)



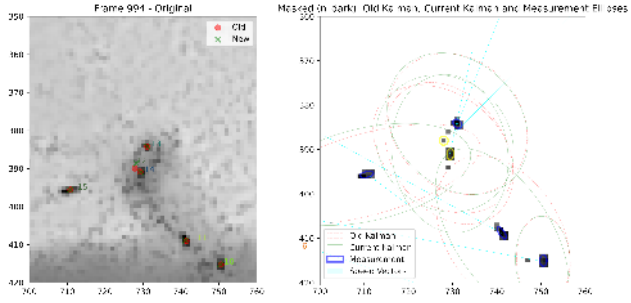
(b) Frame 992 (mod.)



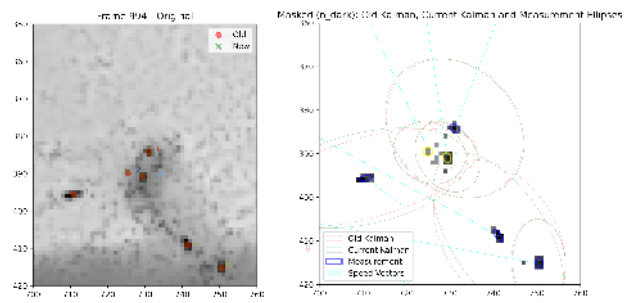
(c) Frame 993 (no mod.)



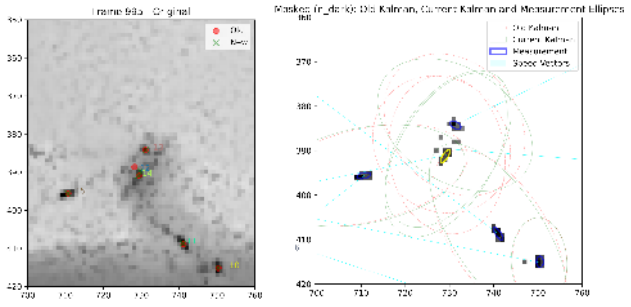
(d) Frame 993 (mod.)



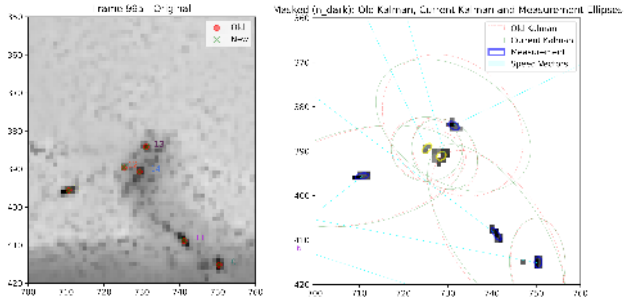
(e) Frame 994 (no mod.)



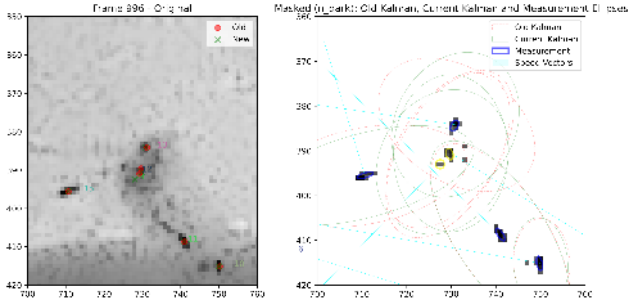
(f) Frame 994 (mod.)



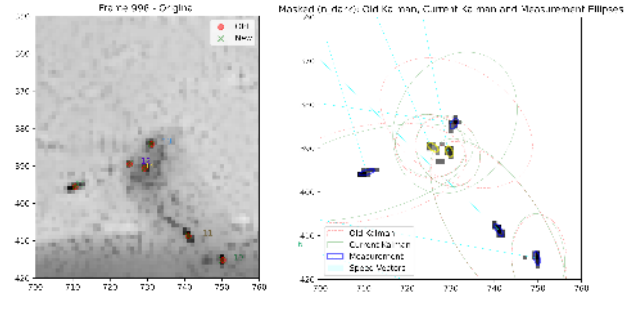
(g) Frame 995 (no mod.)



(h) Frame 995 (mod.)



(i) Frame 996 (no mod.)



(j) Frame 996 (mod.)

Figure 11: Frames 992–996 showing each frame without modulation (left column) and with $\sigma$ reduced by 1000 when the two limb markers are within 5px (right column).

## 6.3 Manual Overrides

Several lateral-view sequences exhibited extremely low contrast and high noise levels, rendering the markers barely distinguishable—even to a trained observer. In those segments, automated tracking consistently failed or produced spurious assignments. To recover reliable annotations, affected frames were manually reviewed and corrected: marker centroids were re-positioned by hand, and track continuity was enforced by interpolating across brief dropouts.

## 6.4 Reprojection

The marker positions obtained from the fluoroscopic views were reprojected onto the optical camera images using a script provided by Chuanfang Ning. This script employs a triangulation-based approach, leveraging calibrated intrinsic and extrinsic camera parameters. Specifically, it uses pairs of fluoroscopic views to triangulate 3D marker coordinates, which are then projected onto the optical camera frames by applying the corresponding camera matrices. This ensures accurate alignment between the fluoroscopic and optical image modalities.

## 6.5 Observed Limitations

- **Per-view parameter-tuning overhead.** Calibrating the tracking script for a single optical view was a labor-intensive, trial-and-error process that typically consumed *an entire working day* ($\sim$8 h). The procedure involved

    1. **Initial marker seeding.** Manually recording the starting $(x, y)$ coordinates for every marker in the first frame.
    2. **Process-noise calibration.** For each functional group of markers, empirically selecting the lowest Kalman $\sigma$ that still allowed the filter to follow rapidly accelerating points *without drifting* (see Figure 12)
    3. **Overlap-penalty tuning.** Identifying pairs of markers that frequently merge and separate, then adjusting their superposition-penalty coefficients $\alpha_{ij}$ until swap errors disappeared.
    4. **Distance constraints (optional).** Adding hard distance bounds for specific pairs whenever it demonstrably improved performance.
    5. **Noise-covariance modulation.** Reviewing the sequence in slow motion, noting frames where two markers swap identities, and manually refining the covariance-modulation parameters described in Section 6.1.4.
    6. **Low-quality sequences.** In the poorest videos, the markers were barely visible; their positions had to be estimated by eye and hard-coded as overrides, further extending the tuning time.

    Because these steps had to be repeated for *every* view and sequence, the cumulative overhead largely erased the intended automation benefit.

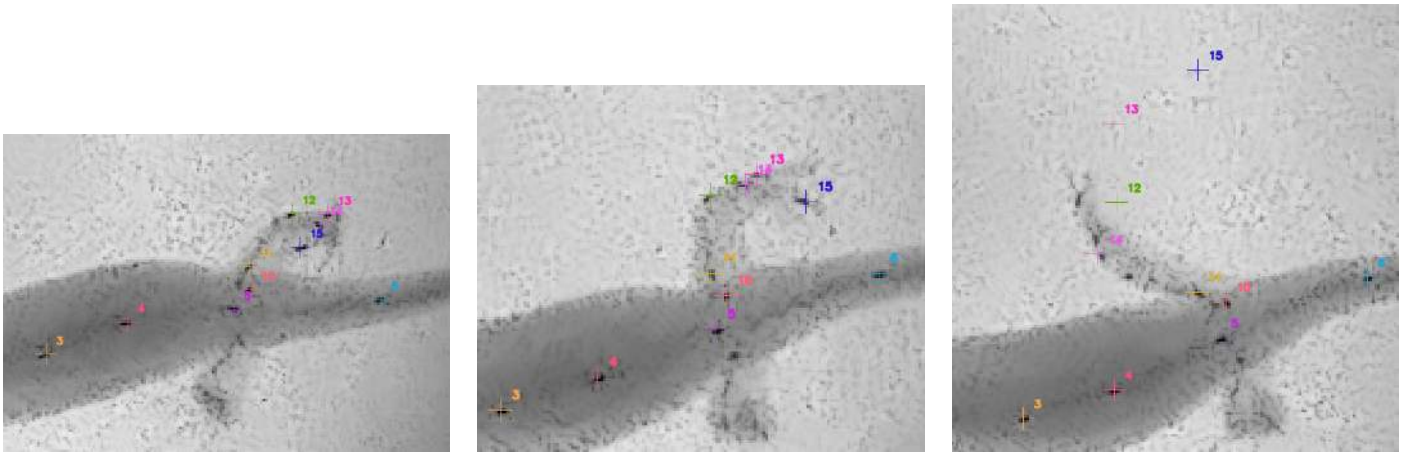
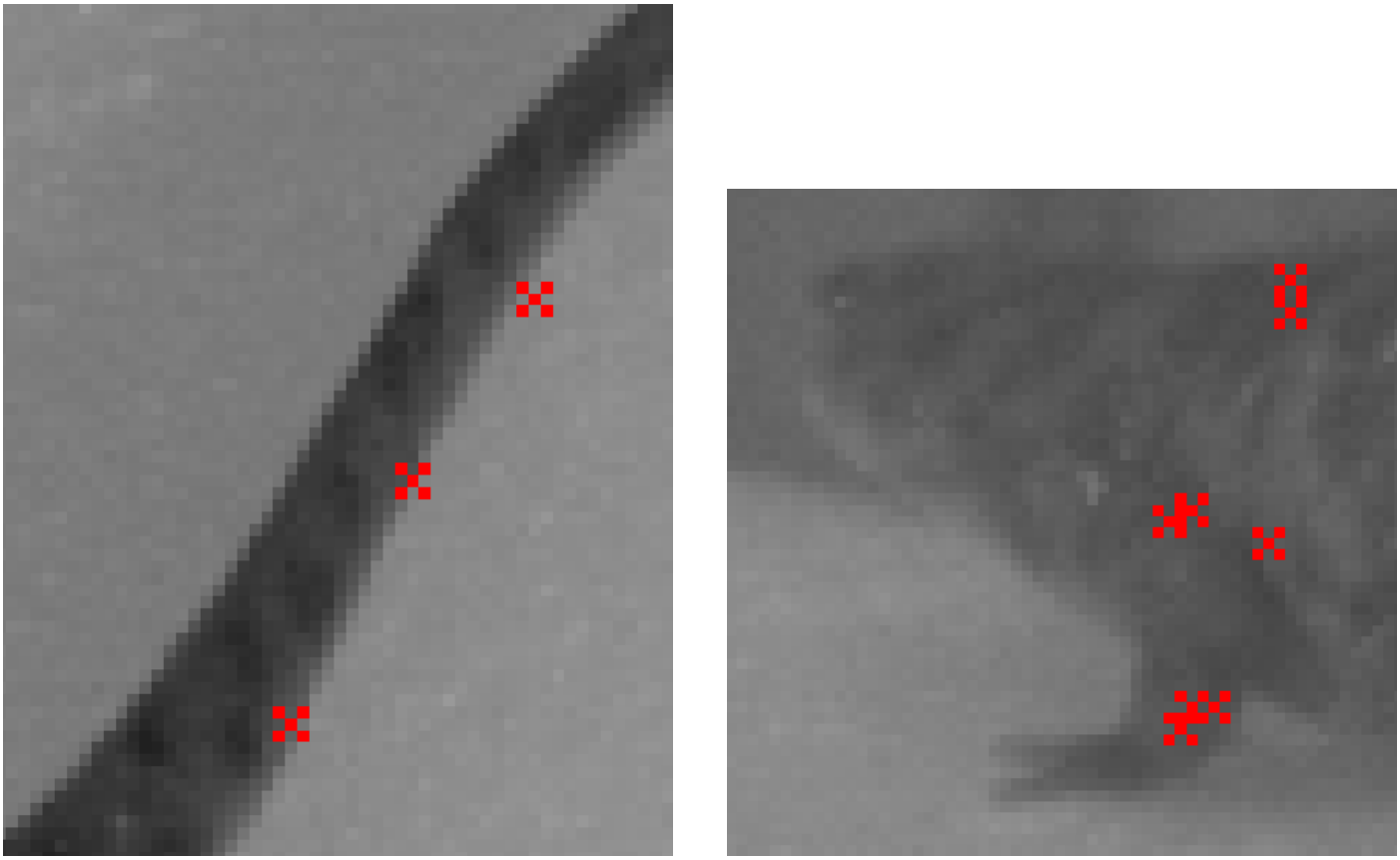


Figure 12: Example of Kalman filter divergence on a fast limb kick when the process-noise parameter $\sigma$ is set too low.

- **Marginal returns from automated labelling.** Although the semi-automatic script can produce thousands of frames after the initial tuning effort, network generalisation benefits primarily from a small, diverse subset of frames per sequence; additional frames are highly redundant and add minimal new information.
- **Reprojection inaccuracies.** Some markers could be placed more precisely by eye than by reprojection, owing probably to calibration error (see Figure 13a).
- **Optical occlusions.** The X–ray inevitably labels body parts even when they are hidden in the optical view, creating misleading supervision for a vision–only network (see Figure 13b).
- **Implant misplacement.** Certain radiopaque beads were implanted at slightly incorrect anatomical locations (e.g., the femur–tibia joint marker placed proximally). When visually evident, these markers had to be manually repositioned to the anatomically correct landmark (see Figure 13b).



(a) Reprojection error on distal caudal vertebrae

(b) Implant misplacement and occluded claw marker

Figure 13: Examples of reprojection and implant placement errors requiring manual correction.

# 7 DeepLabCut training

## 7.1 Dataset

The training dataset consists primarily of annotations derived from Pleurodeles 1 (Sequence 6), which includes three optical views with 20 carefully selected, non-redundant frames per view. Each frame contains annotations of the spine and up to two limbs, subject to their visibility.

To enhance the model's generalisation capabilities, additional manually annotated data were included: two sequences, each recorded from three views with 20 selected frames per view, and one sequence recorded from a single view, also containing 20 frames. In these additional sequences, limb annotations were performed manually for every selected frame. However, spine annotations were only provided for a single frame per sequence due to time constraints, thus ensuring minimal representation of spinal structure across these sequences. Examples of such manual annotations are shown in Figures 14d and 14e.



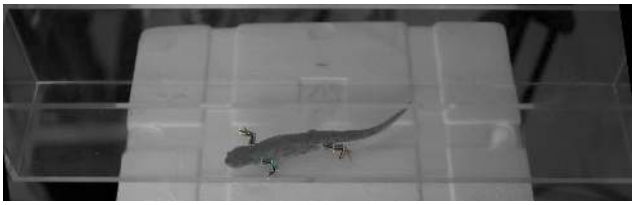
(a) X-ray View 1



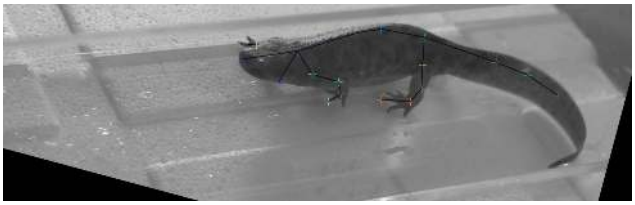
(b) X-ray View 2



(c) X-ray View 3



(d) Manual (limbs only)



(e) Manual (limbs + spine)

Figure 14: Top row: Examples of annotated frames from the three optical views. Bottom row: Examples from manually annotated sequences, showing a frame with only limbs (left) and a frame where both limbs and spine were annotated (right).

## 7.2 Symmetry Augmentation

Since not all anatomical landmarks are always visible or annotated, the model training accommodates partial annotations. To explicitly leverage the bilateral symmetry of the salamander, the dataset was augmented by horizontally mirroring both images and their corresponding labels. During this process, landmark labels corresponding to left or right limbs were appropriately inverted (e.g., the left front girdle annotation becomes the right front girdle annotation). This symmetry augmentation ensures that the model robustly tracks anatomical landmarks on both sides of the animal. The whole dataset with the symmetry augmentations contained 376 frames.

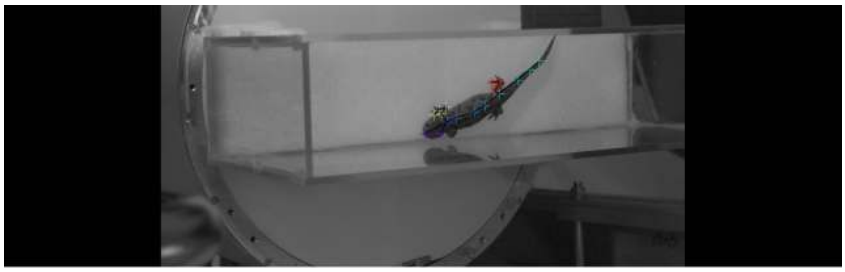


Figure 15: Example of symmetry augmentation: the image is the horizontal mirror of Figure 17a), with left and right landmark annotations swapped accordingly.

## 7.3 Cropping

To increase training speed the images were cropped from 1280x800 to 1280x400 and centered on the animal.

## 7.4 Hyper-parameter Selection and Library Adjustments

All experiments were executed on the EPFL *Izar* cluster with a single NVIDIA V100 PCIe (32 GB) GPU and eight CPU cores. The mini-batch size was gradually increased until the GPU utilisation remained above 90 %; a value of **8** achieved full saturation without exhausting the 32 GB of device memory.

**CPU input bottleneck.** The TensorFlow branch of DeepLabCut originally spawns only *one* data-loading thread, which left seven CPU cores idle and throttled the input pipeline. A minor code modification in the DLC library enabled multi-threaded prefetching, eliminating the CPU bottleneck.

**Training runs.** Both experiments were trained with **Adam**, differing only in the way the learning-rate (LR) is staged. For clarity, each `multi_step` entry is to be read as *"use this LR for all iterations **below** the stated milestone until a new value overrides it."*

1. **Schedule A (4-step).**

```
multi_step:
  - [0.0001,   10000]   # 0 to 9,999 it.
  - [0.00001,  50000]   # 10,000 to 49,999 it.
  - [0.000001,100000]   # 50,000 to 99,999 it.
  - [0.0000001,300000]  # 100,000 to 300,000 it.
```

2. **Schedule B (6-step, warm-up).**

```
multi_step:
  - [0.00001,   500]    # 0 to 499 it.    (warm-up)
  - [0.00005,  1000]    # 500 to 999 it.
  - [0.0001,   8000]    # 1,000 to 7,999 it.
  - [0.00002, 16000]    # 8,000 to 15,999 it.
  - [0.000005,28000]    # 16,000 to 27,999 it.
  - [0.000001,40000]    # 28,000 to 40,000 it.
```

Schedule B introduces a short 500-iteration warm-up and a sharper LR decay up to 40 k iterations, whereas Schedule A follows a coarser four-stage decay extending to 300 k iterations.

# 8 Results

## 8.1 Evaluation Metrics

**Pixel Error (PE).** We report the *mean Euclidean distance*, in pixels, between each predicted landmark and its ground–truth coordinate. Two variants are distinguished:

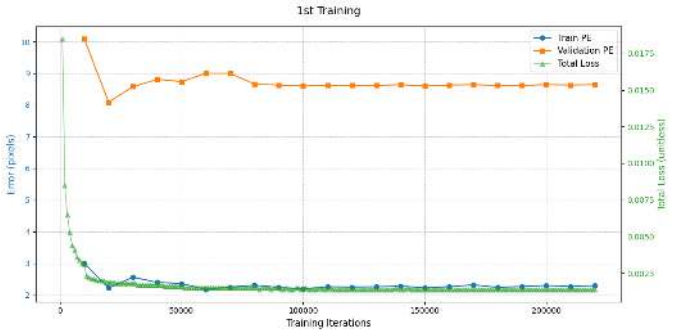
- **Training Pixel Error (Train PE)** — averaged over all frames in the training set.
- **Validation Pixel Error (Val PE)** — computed on a held-out validation set and therefore indicative of generalisation.

**Total Loss.** During optimisation the network minimises a dimension-less *Total Loss*, defined as the sum of pixel-wise cross-entropy terms over all score-maps (plus any regularisation). Whereas Train PE and Val PE measure geometric accuracy in the image plane, Total Loss quantifies how well the predicted probability distributions match the ground-truth heat-maps.

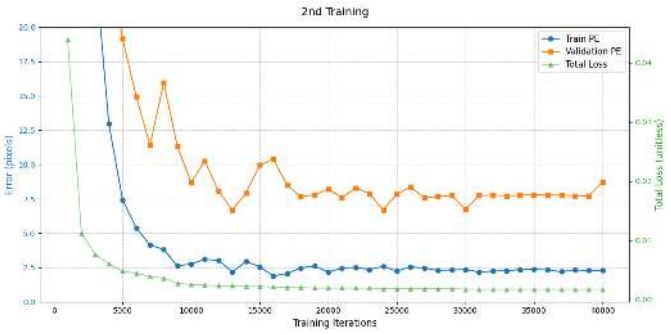
## 8.2 Training Histories

Figure 16 contrasts two training schedules:

1. **Run A** (default decay). Learning rate milestones were spaced sparsely; metrics were recorded every 10 000 iterations. After $\sim$ 20 000 iterations the Val PE rises again while Train PE continues to decrease, signalling over-fitting.
2. **Run B** (accelerated decay, dense snapshots). The learning rate is reduced more aggressively and all metrics are logged every 1 000 iterations. The finer sampling reveals a lower Val PE plateau reached earlier, preventing unnecessary computation and mitigating over-fitting.




(a) Run A: sparse milestones; over-fit beyond 20 k iter.




(b) Run B: faster decay; dense snapshots (1 k iter).

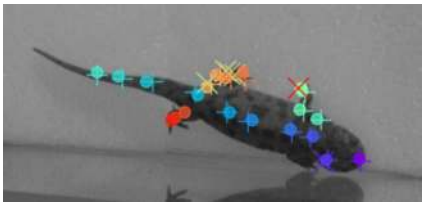
Figure 16: Training (blue), validation (orange) pixel errors and Total Loss (green, right axis) for the two learning-rate schedules.

Run B attains its minimum Val PE in roughly one-third of the iterations required by Run A, demonstrating that an earlier, sharper learning-rate decay better matches the convergence speed of the network on this dataset. The chosen model snapshot to discuss the results was the 13000th iteration of Run B, where VAL PE is the lowest.

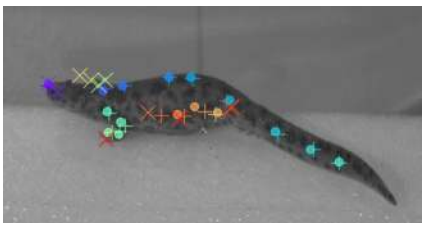
## 8.3 Validation Set Results

In the panels below, **plus signs** denote ground-truth landmarks, **circles** mark predictions whose confidence exceeds 60 %, and **crosses** indicate lower-confidence estimates. All frames belong to the *validation set* and were never seen during training. The first row stems from the fully annotated X-ray–guided dataset, whereas the second row shows manually labelled sequences in which only the limb landmarks were provided.

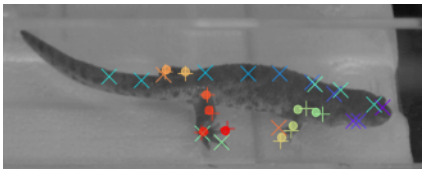
Two trends emerge. (i) When a limb is occluded on the far side of the animal, the network often "projects" the invisible joints onto the visible ones, assigning them low confidence (crosses), while the visible joints remain high-confidence circles. (ii) For the manually annotated sequences the spine is always predicted with low confidence, reflecting the limited number of spinal annotations available during training; importantly, a confidence threshold that accepts these uncertain spine estimates while still rejecting the spurious limb projections doesn't exist.



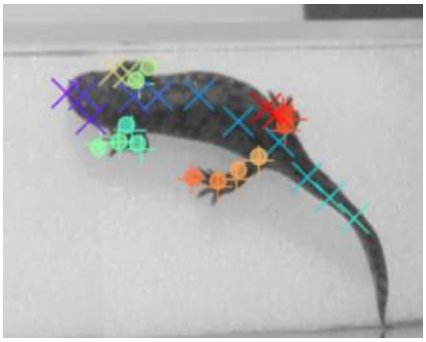
(a)



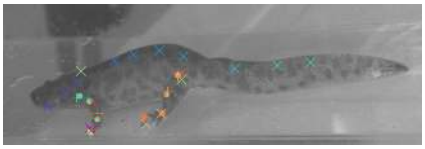
(b)



(c)



(d)



(e)

Figure 17: Validation examples: fully annotated XR-guided frames (top row) versus sparsely annotated manual frames (bottom row).

The left–right mirroring augmentation proved only partly effective. In panel 17a the network correctly localises the *left hind limb*, even though this limb was never annotated in that sequence, but not the left front limb. The bilateral *head* markers often collapse onto the same location, indicating that the model has not learned to disambiguate them.

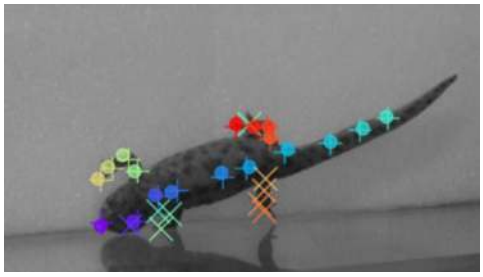
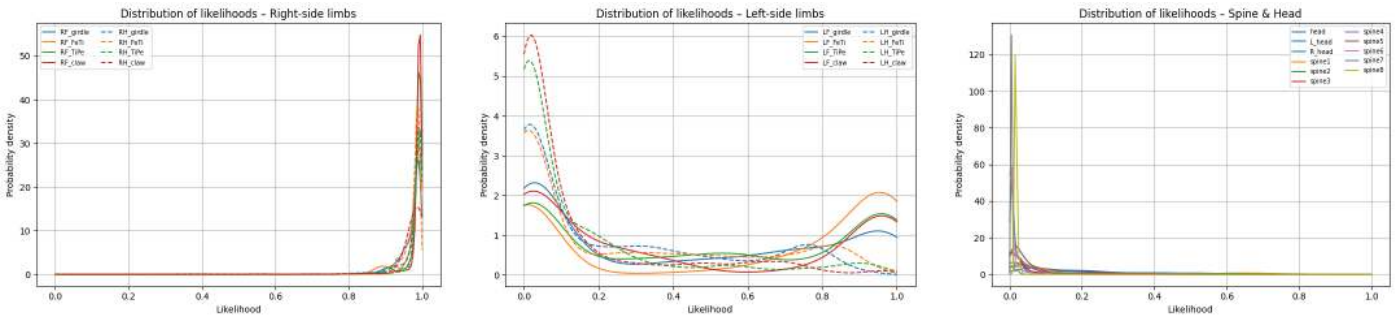


Figure 18: Training-set example illustrating symmetry augmentation: the network predicts both fore- and hind-limb joints on the unannotated side (circles), although only the opposite side was labelled during training.

## 8.4 Test Results – Complex-Terrain Sequence

The best network was challenged with a complete walking sequence on uneven ground that had been *entirely excluded* from both training and validation. Two complementary evaluations are reported:

1. likelihood histograms for the main anatomical families, and
2. qualitative inspection of four consecutive frames rendered with a global threshold $T = 0.6$, followed by two frames rendered with a mixed strategy ($T = 0.5$ for limbs, no threshold for axial landmarks).




(a) Right-side limbs (b) Left-side limbs (c) Spine & head

Figure 19: Normalised likelihood distributions on the complex-terrain sequence ($N = 4\,000$ frames). The dashed line marks the working threshold $T = 0.6$.

**Likelihood histograms.** **Right limbs** (Fig. 19a) show a single, sharp peak near 1.0, confirming the model's high confidence whenever those limbs are in direct view.

**Left limbs** (Fig. 19b) are distinctly bimodal: a high-confidence mode (0.9) when unobstructed and a low-confidence mode (0.1) under self-occlusion. The rear limb accumulates slightly more mass in the low mode, matching its more frequent occlusion by the body.

**Spine and head** (Fig. 19c) almost never rise above 0.1. These landmarks are severely under-represented in the manual subset of the training set; the network therefore receives too little positive signal to form reliable heat-maps.

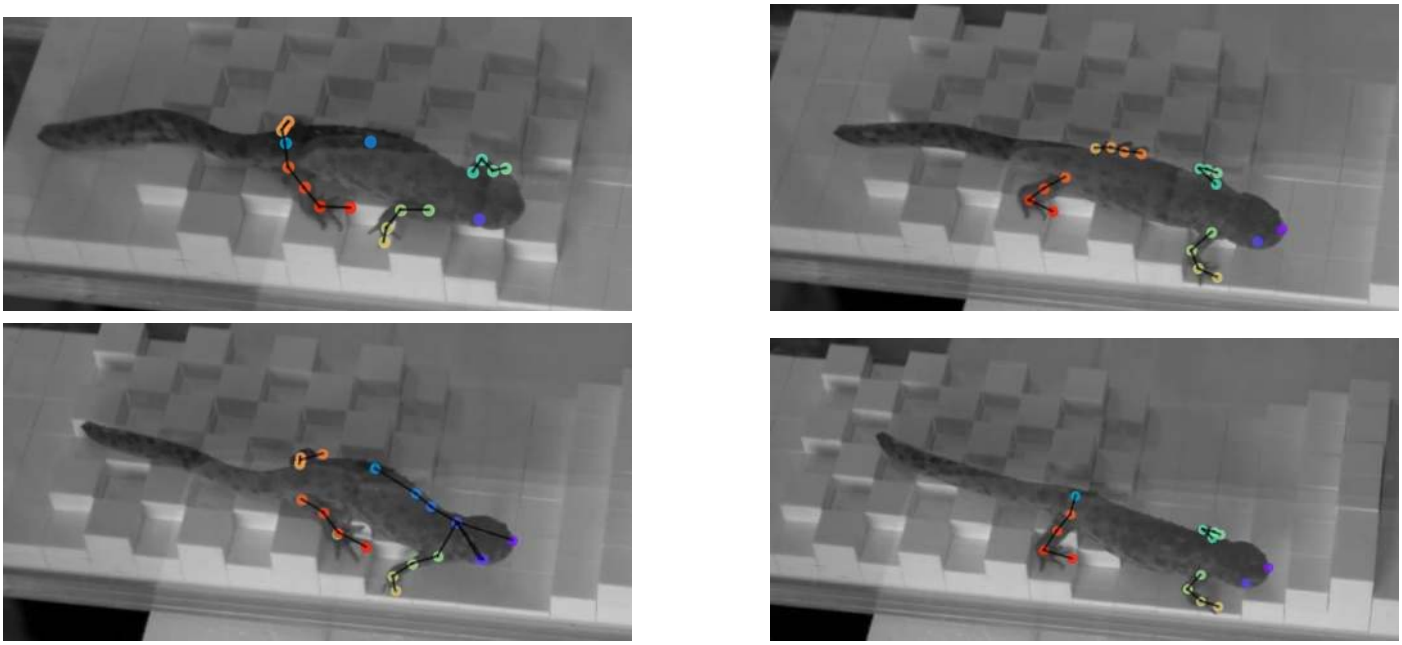


Figure 20: Four consecutive frames rendered with a global likelihood threshold $T = 0.6$. Invisible or low-confidence landmarks are transparent.

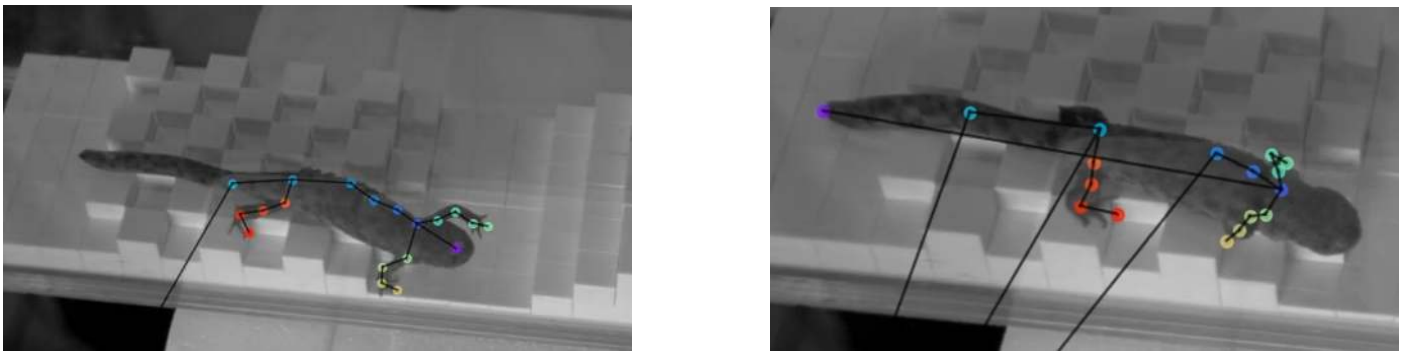


Figure 21: Forcing all axial landmarks to display reveals a mixture of accurate predictions and evident mis-placements on the spine.

**Qualitative examples.** **Discussion.** Lowering the threshold selectively for spine and head does surface a few correct detections (Fig. 21), but also admits many false positives. A dynamic, part-dependent threshold therefore offers marginal benefit. The remedy is not a more elaborate thresholding rule but a *better-balanced training set*—either by adding spine/head annotations or by raising their loss weight during training.

# 9 Conclusions and Future Improvements

The prototype pipeline can be strengthened on four fronts—data acquisition, image pre-processing, semi-automatic tracking, and deep-learning training. For each domain we summarise the present shortcomings and the most direct corrective actions.

## 9.1 Data Acquisition (X-ray & Optical)

**Current limitations.** The raw X-ray footage is noisy, and several markers become indistinct whenever their projections overlap with bone or with neighbouring beads, resulting in poor local contrast.

**Suggested improvements.**

- Record at higher detector resolution to increase marker-to-background contrast.

- Re-position the imaging planes (view angles) to minimise persistent superpositions.

## 9.2 Image Pre-processing

**Current limitations.** Contrast-enhancement and denoising parameters are adjusted manually for every recording; the overall pipeline is therefore sub-optimal and time-consuming.

**Suggested improvements.**

- Refine the pre-processing chain and automate parameter selection.

## 9.3 Semi-Automatic Tracking Script

**Current limitations.** Each new sequence requires roughly one working day of manual tuning: initial marker positions, group-specific process-noise levels, superposition penalties, distance constraints, and noise-covariance modulation are all set by hand. Frames with very low contrast still need manual overrides.

**Suggested improvements.**

- Replace separate 2-D Kalman filters with a single 3-D Kalman model that integrates both views simultaneously.
- Apply sensible default hyper-parameters to all markers before undertaking fine per-marker adjustments.

## 9.4 Deep-Learning Model Training

**Current limitations.** Because of software constraints on the *Izar* cluster, training was conducted with `DeepLabCut` 2.3.1 under TensorFlow. This legacy version is limited to a single GPU and lacks recent features; TensorFlow support is also being phased out.

**Suggested improvements.**

- Upgrade to the latest PyTorch-based version of DeepLabCut, which restores multi-GPU capability and introduces newer functionalities.